Checkpoint 5 — k-means Objective & PCA

INFO 521 — Machine Learning Foundations

> ℹ Course-schedule placement
>
> **Weeks 6–7** · with **Modules 6–7 — Clustering & PCA** · gates **Project 2.3 / Gate B**.

Gates: Project 2.3 (Clustering) — and with it, Gate B. **Format:** in-class, closed-book, ~15–20 min, one derivation/explanation. Graded Satisfactory / Not-Yet. **Retakeable** (an alternate version is provided below).

This checkpoint carries the unsupervised-gate recall outcomes: the k-means objective and why model selection for K is non-trivial (Version A), and the PCA procedure and the meaning of its eigenstructure (Version B).

Prompt (Version A)

State the **k-means objective** J as a function of the cluster assignments and centroids. Explain why choosing K by **minimizing J over K** trivially fails, and describe **one principled method** for selecting K.

Prompt (Version B — retake)

Describe the **PCA procedure** step by step for a centered data matrix, and state what the **eigenvalues** and **eigenvectors** of the covariance matrix represent.

Satisfactory criteria

- (A) Correct objective J (within-cluster sum of squared distances); states that J is **non-increasing in K** and hits 0 at $K = N$, so minimizing J always prefers more clusters; names a valid criterion (elbow, silhouette, or gap) and describes it.
- (B) Lists center → covariance → eigendecompose (or SVD) → sort → project; states eigenvectors = principal directions (orthogonal axes of maximal variance), eigenvalues = variance along each, and the ratio $\lambda_i / \sum_j \lambda_j$ = proportion of variance explained.

Answer key

A: With assignments $z_n \in \{1, \ldots, K\}$ and centroids $\boldsymbol{\mu}_k$,

$$J = \sum_{n=1}^{N} \left\| \mathbf{x}_n - \boldsymbol{\mu}_{z_n} \right\|^2 = \sum_{k=1}^{K} \sum_{n \in C_k} \left\| \mathbf{x}_n - \boldsymbol{\mu}_k \right\|^2 .$$

J can only **decrease** as K grows (more centroids ⇒ every point can sit nearer one), reaching $J = 0$ when $K = N$ (each point is its own cluster). So "minimize J over K" degenerately selects $K =$

N and is useless as a selection rule. Pick K instead by, e.g., the **elbow** (the K past which J falls only marginally) or the **silhouette** (choose the K maximizing the mean silhouette $s_n = (b_n - a_n)/\max(a_n, b_n)$, where a_n/b_n are the mean intra-cluster and nearest-other-cluster distances).

B: (1) **Center** the data, $\mathbf{X}_c = \mathbf{X} - \bar{\mathbf{x}}$ (standardize if features differ in scale). (2) Form the **covariance** $\mathbf{\Sigma} = \frac{1}{N-1}\mathbf{X}_c^\top \mathbf{X}_c$. (3) **Eigendecompose** $\mathbf{\Sigma} = \mathbf{V}\mathbf{\Lambda}\mathbf{V}^\top$ (or take the SVD $\mathbf{X}_c = \mathbf{U}\mathbf{S}\mathbf{V}^\top$, with $\lambda_i = s_i^2/(N-1)$). (4) **Sort** components by descending eigenvalue. (5) **Project** onto the top k: $\mathbf{Z} = \mathbf{X}_c\mathbf{V}_k$. The **eigenvectors** are the principal directions — orthogonal axes of greatest variance; each **eigenvalue** λ_i is the variance captured along its direction, and $\lambda_i / \sum_j \lambda_j$ is the proportion of total variance that component explains.